Vedant Gupta

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.18
Bal Bharati Public School, Bhopal	Class XII (CBSE)	2023	93.40%
Bal Bharati Public School, Bhopal	Class X (CBSE)	2021	94.20%

SCHOLASTIC ACHIEVEMENTS

- Cleared NTSE stage 1 with **state Rank 120** among 2.5+ lakh candidates
- Selected for Amazon Summer School 2025, ranked among the **top 3,000** students in India

SKILLS

Languages: Python, C++, C, SQL

Deep Learning & ML: PyTorch, TensorFlow, Hugging Face, Transformers, Vision Transformers, Cross-Encoders, Contrastive Learning

Agentic AI & LLMs: Agentic Workflows, MCP Server, Prompt Engineering, Multimodal RAG, Vision-Language Models, LLM-as-a-Judge

CS Fundamentals: DSA **(Codeforces: 1400+, LeetCode: 1600+, 500+ Problems Solved)**, OOP, OS, DBMS, System Design

WORK EXPERIENCE

Associate Machine Learning Intern, ServiceNow

- Fine-tuned a **Harrier-270M retriever and BGE cross-encoder** using a custom hard-negative dataset, boosting **nDCG@10 by 7.9%** and matching an 8B-parameter SOTA reranker at 1/32nd the size.
- Curated a standardized evaluation dataset combining public and production data to benchmark retrieval, reranking, and multimodal RAG architectures.
- Designed a **Multimodal RAG proof-of-concept** to upgrade the retrieval pipeline using ColPali, **improving Recall@1 by 12%** over legacy OCR, captioning, and text embedding methods.
- Built MCP server to automate model deployment, evaluation and testing pipelines across instances, eliminating manual overhead.

PUBLICATIONS AND APPLIED RESEARCH

Detecting Synthetic Images with Frozen DINOv3 Layer Features

- Developed **a synthetic image classifier** using a custom 3-layer Transformer head trained on frozen DINOv3 ViT-L tokens.
- Executed a layer-depth ablation study, proving deeper network **layers (L12+L24)** provide the strongest discriminative signal.
- Achieved **0.902 F1 and 0.862 recall** across GAN and latent-diffusion datasets on the open-run validation set.

RESOLVENOW: Rule-Based Type Classification & LLM-Driven Tagging

- Designed a single-call prompt architecture using checklist scaffolding with DeepSeek-R1-8B to execute multi-label tagging across a 141-code taxonomy.
- Engineered a French-English lexical classifier for UN resolutions, achieving **a 0.910 F1** for paragraph-level type classification.

- **Secured 7th place overall and ranked 2nd globally** for reasoning quality, evaluated by an open-weight LLM-as-Judge.

PROJECTS

Event-Driven Datastore with Custom Thread Scheduler

- Engineered an event-driven, Redis-compatible in-memory datastore from scratch in **C++20**, implementing a custom user-level thread scheduler to bypass traditional kernel scheduling overhead.
- Validated system performance against the official redis-benchmark suite, maintaining **~90,000 RPS with ~10ms p50 latency** for 1,000 concurrent connections.

Hangman Solver: Phase-Specialized Transformers & Meta-Learning

- Implemented a **RoPE-based Transformer** to play Hangman via Masked Language Modeling without relying on dictionary lookups.
- Trained phase-specialized Transformer experts (Opener, Generalist, Closer) using a hybrid sampling pipeline, mitigating exposure bias and reducing the train-inference gap.
- Built a Meta-Learner on experts, merging predictions via Soft-Voting + Noisy-OR gate for a **72.04% win rate** on 250k unseen word

Lossless Huffman Compressor

- Implemented a Huffman Coding algorithm in C++ using std::priority_queue (min-heap) to generate optimal variable-length prefix trees from character frequencies.
- Developed modular encoder, decoder, and verification tools to handle binary file serialization, metadata parsing, and byte-level lossless data validation.

POSITIONS OF RESPONSIBILITY

Vice Overall Coordinator, E-Cell IIT Hyderabad

- Directed a 60-member team executing flagship events E-Summit and E-Merge, engaging 2,000+ attendees, 15+ startups, and VCs like Draper and Unicorn India Ventures.
- Spearheaded 'Startup Studio' product strategy and development, connecting early-stage founders to investors via the Startup Fair and Fetching Fortunes x Ideabaaz, securing national Zee TV features for top startups.

EXTRACURRICULAR

Represented India in **Japan under the *JENESYS* Student Cultural Exchange Program**, selected among the top 24 of 1000+ applicants

Promoted and actively participated in Photography Club activities as Infocus Coordinator in 2024